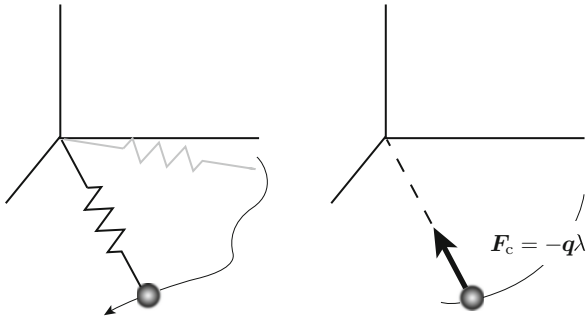


Fig. 4.4 *Left*: a flexible pendulum model; the bob moves on a path under some applied force defined by the potential U and the stiff spring holding the bob near a fixed distance from the origin. *Right*: a constrained version of the same model, where the spring has been replaced with a rod; the tension in the rod acts on the bob so as to maintain a precise fixed distance from the origin

To formulate a constrained model we use Gauss's principle of least constraint which says that the force $\boldsymbol{F}_{\mathrm{c}}$ needed to compensate for the presence of a constraint of the form

$$g(\boldsymbol{q}) \stackrel{\text{def}}{=} \|\boldsymbol{q}\|^2 - r_0^2 = 0$$

will be transverse to the tangent space of the manifold $\{\boldsymbol{q} | g(\boldsymbol{q}) = 0\}$, that is, it must point in the normal direction

$$\boldsymbol{F}_{\mathrm{c}} = -\boldsymbol{q}\lambda,$$

where $\lambda \in \mathbb{R}$ will vary over time so that the force $\boldsymbol{F}_{\mathrm{c}}$ represents the tension in the stiff rod holding the pendulum bob to the origin (right panel of Fig. 4.4) The equations of motion for the constrained system take the form

$$\begin{aligned}
\dot{\boldsymbol{q}} &= \boldsymbol{v}, \\
\dot{\boldsymbol{v}} &= -\nabla U(\boldsymbol{q}) - \boldsymbol{q}\lambda, \\
0 &= \|\boldsymbol{q}\|^2 - r_0^2.
\end{aligned}$$

These are called the Euler-Lagrange equations, and they may be derived formally by use of the calculus of variations. We refer to λ, which is a function of time, as a Lagrange multiplier.

More generally, if we have a system with N_c configurational coordinates[1] and l constraints $g_1(\boldsymbol{q}) = 0, g_2(\boldsymbol{q}) = 0, \ldots, g_l(\boldsymbol{q}) = 0$, the tension forces due to the

[1]Note that with l constraints the actual number of degrees of freedom, which is needed for things like the computation of temperature, is $\mathrm{N}_{\mathrm{d}} = \mathrm{N}_c - l$.